MICROSOFT CERTIFICATION SERIES

AI-901: Azure AI Fundamentals

Module 1&2

Fundamental AI Concepts

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ARTIFICIAL INTELLIGENCE & MACHINE LEARNING WITH TOUSEEF AHMED

PMI	Microsoft	Google	University of Oxford	Harvard University	coursera
PMP Certified	5x Certified	AI & Data Analytics	AI • Law & Justice	Machine Learning	Multiple Certs

100+	~80,000	~5,000	1,200+	~1,000
International Certifications	Professionals Trained	Intl. IT Certifications Facilitated	Microsoft-Certified (CMSDI)	Employment Opportunities Generated

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING
WITH
TOUSEEF AHMED

Project Management Institute (PMI): PMP – Project Management Professional Microsoft (5 Associate/Leader-level certifications): AI-300 Machine Learning Operations Engineer Associate · AI-102 Azure AI Engineer Associate · AB-731 AI Transformation Leader · PL-300 Power BI Data Analyst Associate · DB-750 Azure Databricks Data Engineer Associate Google: Google AI Certified · Google Data Analytics Certified Oxford University: Certified in Artificial Intelligence · Certified in Law & Justice Harvard University: Certified in Machine Learning Coursera: Multiple professional certifications across AI, ML, and Data Science Additional Platforms: 70+ additional certifications across AI, ML, and Data Science tracks

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Azure -901

Fundamentals of AI

Machine learning

Computer vision

CERTIFICATION :

- **AI-900 Microsoft Azure**
- https://youtu.be/3Arj5zlUPG4
- Product of Microsoft
- Launched on 1^{st} Feb 2010.
- **AI-901 Microsoft Azure (30 June)**

AI-901 MICROSOFT AZURE

- Cloud computing platform and an online portal to access and manage resources and services provided by Microsoft
- 80 % of Fortune companies using services
- Has 42 data centers

(Competitors of Azure)

AZURE SERVICES:

- More than 18 categories and contain more than 200 services
- (categories name)

USES OF AZURE

- Application development, testing, application hosting, creating virtual machines, integration and syncing devices, collecting and storing metrics, virtual hard drives,

What Is Artificial Intelligence?

Artificial Intelligence (AI) refers to software systems that imitate human-like capabilities such as perception/understand, reasoning/result, learning, and decision-making. On Azure, AI is delivered through pre-built services and Microsoft Foundry, the unified platform for building, deploying, and managing AI models and intelligent agents.

5
Core AI workload areas

6
Responsible AI principles

1
Unified platform: Microsoft Foundry

Why AI Matters on Azure

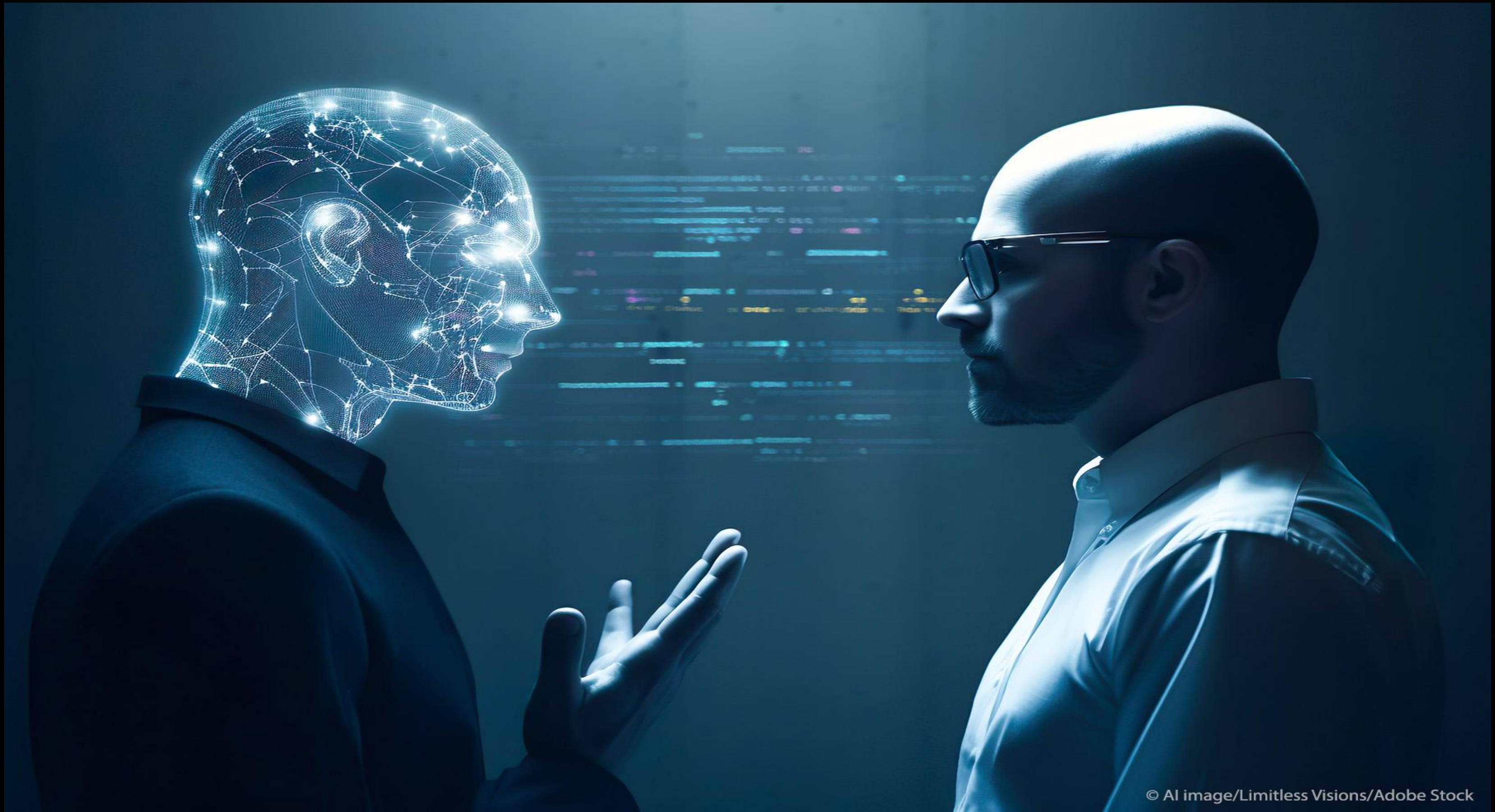
- Turns raw data into predictions, insights, and automated action
- Powers generative and agentic AI experiences across products
- Backed by Azure's global, secure, and scalable infrastructure
- Enables developers to build responsibly using governed tools

FUNDAMENTALS OF AI :

- AI enables us to build amazing software that can improve health care, enable people to overcome physical disadvantages, empower smart infrastructure, create incredible entertainment experiences, and even save the planet!

INTRODUCTION TO AI :

AI is software that imitates human behaviors and capabilities.



INTRODUCTION TO AI :

INTRODUCTION TO AI :

Key workloads include:

- Machine learning
- Computer vision
- Natural language processing
- Document intelligence
- Knowledge minning
- Generative AI

UNDERSTANDING MACHINE LANGUAGE:

- machine learning, a branch of AI that combines computer science and mathematics.

FUNDAMENTAL AI CONCEPTS

Machine Learning: The Foundation of AI

Machine learning (ML) is the process of training a model on data so it can make predictions or decisions without being explicitly programmed for every scenario. Azure supports the full ML lifecycle: data preparation, training, evaluation, and deployment.

01

Data Collection

Gather and prepare representative, labeled data.

02

Model Training

Algorithms learn patterns and relationships in the data.

03

Evaluation

Test accuracy, precision, and reliability on new data.

04

Deployment

Publish the model as a service consumed by applications.

Three common ML approaches: Regression (predicting a number), Classification (predicting a category), and Clustering (grouping similar data points without labels).

UNDERSTANDING MACHINE LANGUAGE:

MACHINE LEARNING WORKING:

- The answer is, from data.
- Data scientists can use all of that data to train machine learning models that can make predictions and inferences based on the relationships they find in the data.

MACHINE LEARNING IN MICROSOFT AZURE

- **Azure Machine Learning** service - a cloud-based platform for creating, managing, and publishing machine learning models

AZURE MACHINE LEARNING STUDIO OFFERS MULTIPLE AUTHORING EXPERIENCES SUCH AS:

- Automated machine learning:
- Azure Machine Learning designer:
- Data metric visualization:
- Notebooks: (Jupyter Notebook servers

UNDERSTAND COMPUTER VISION:

- Computer Vision is an area of AI that deals with visual processing.
- The **Seeing AI** app is a great example of the power of computer vision. (VIDEO)

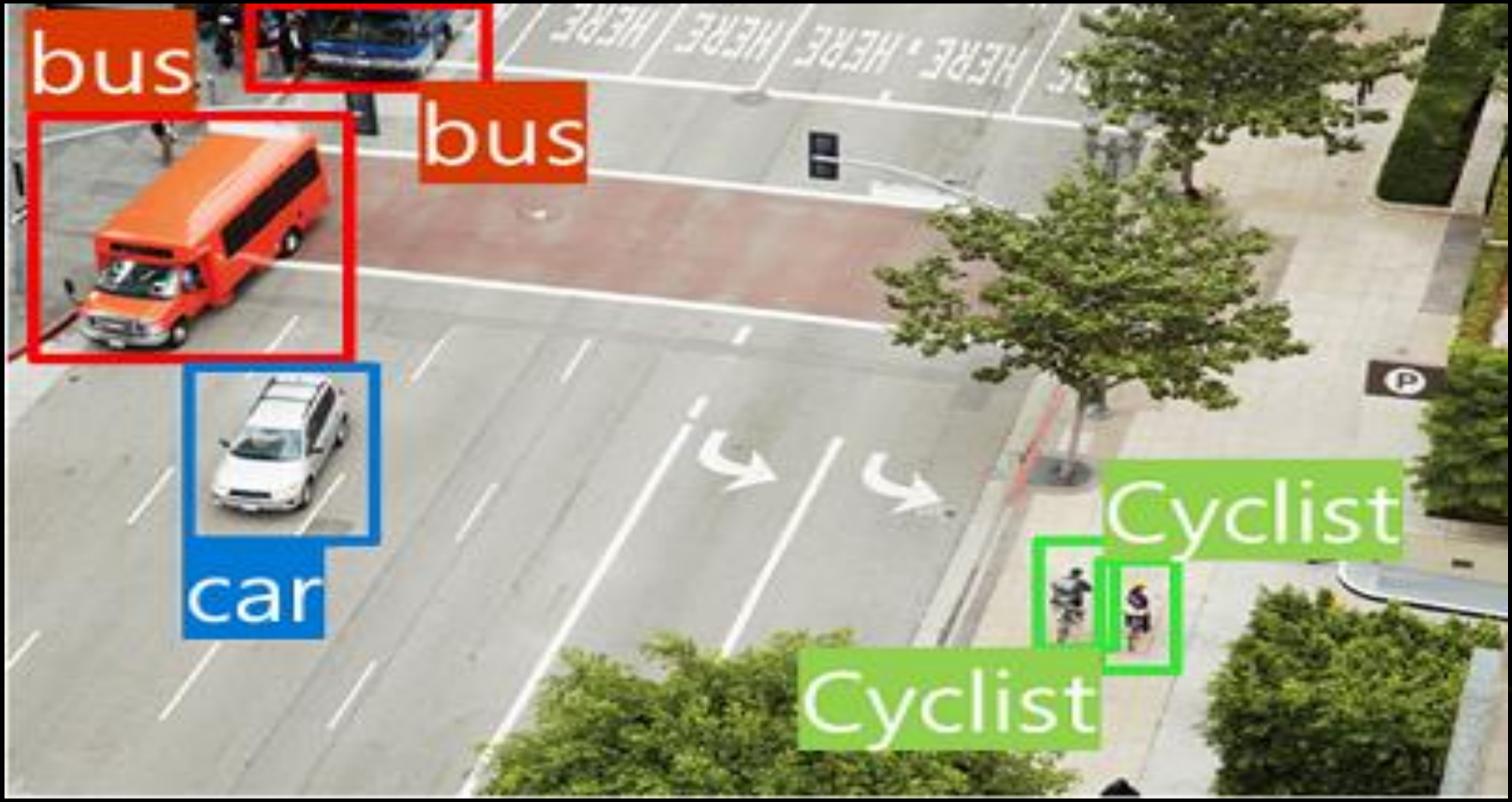
COMPUTER VISION

COMPUTER VISION MODELS AND CAPABILITIES

- **Image classification**

COMPUTER VISION MODELS AND CAPABILITIES

- Object detection



COMPUTER VISION MODELS AND CAPABILITIES

- Semantic segmentation

COMPUTER VISION MODELS AND CAPABILITIES

- Image analysis

COMPUTER VISION MODELS AND CAPABILITIES

- Face detection, analysis, and recognition

COMPUTER VISION MODELS AND CAPABILITIES

- Optical character recognition (OCR)

COMPUTER VISION SERVICES IN MICROSOFT AZURE

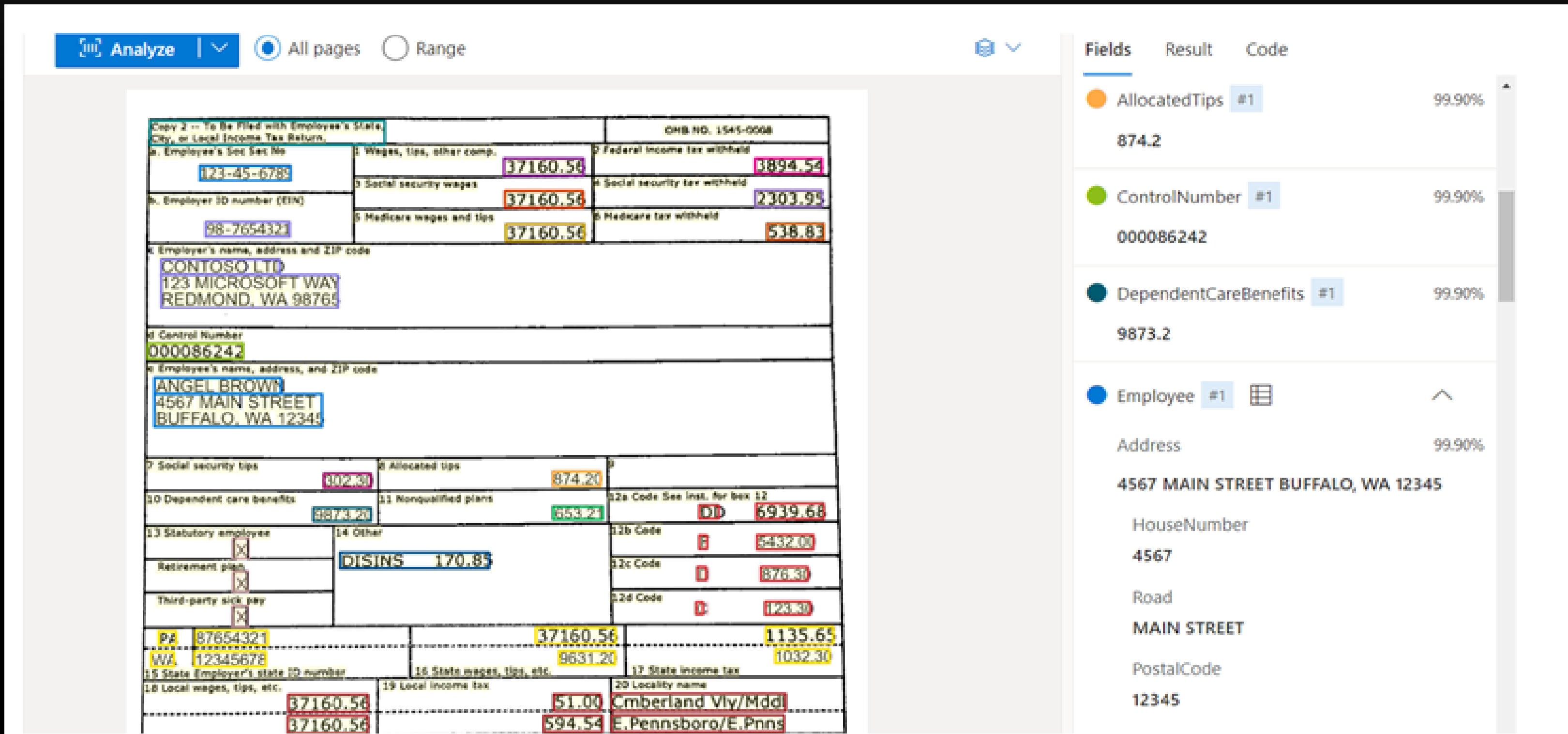
- Microsoft's Azure AI Vision to develop computer vision solutions
- **Image Analysis**:
- **Face**:
- **Optical Character Recognition (OCR)**

INTRODUCTION TO NLP

- Analyze and interpret text in documents, email messages, and other sources.
- Interpret spoken language, and synthesize speech responses.
- Automatically translate spoken or written phrases between languages.
- Interpret commands and determine appropriate actions.

INTRODUCTION TO NLP

DOCUMENT INTELLIGENCE IN MICROSOFT AZURE

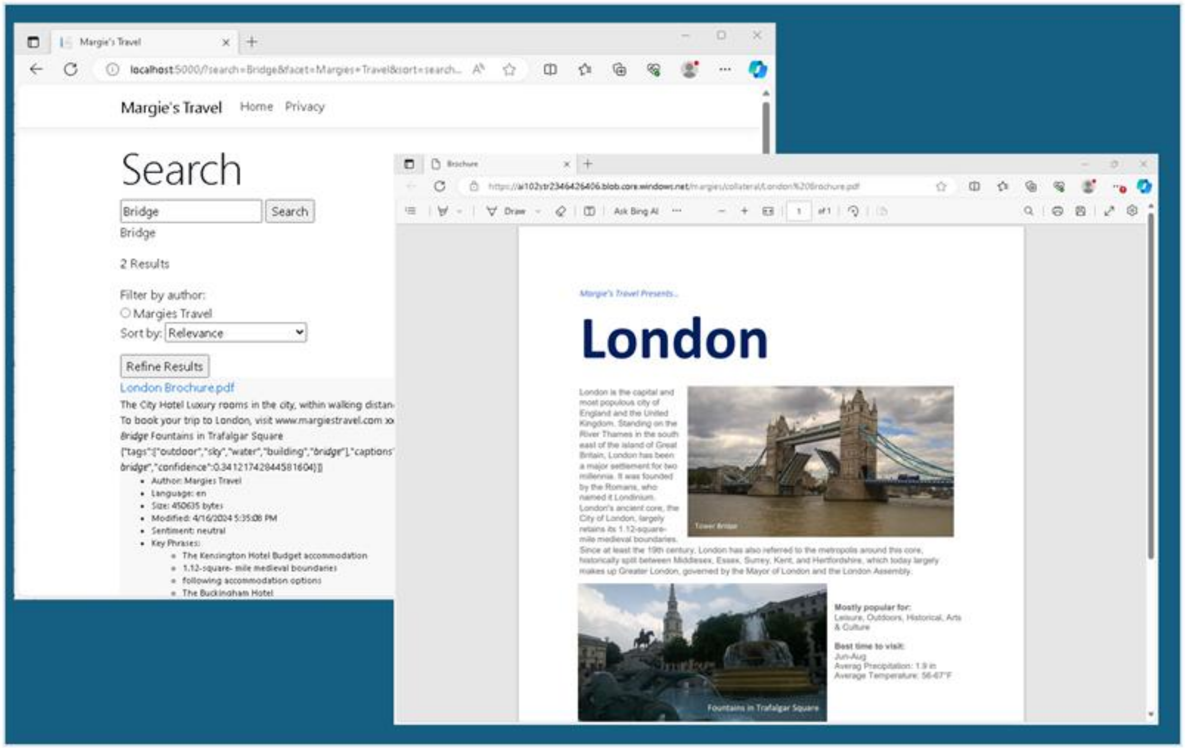


This example shows information extracted from a tax form using Azure AI Document Intelligence.

KNOWLEDGE MINING

- **Knowledge mining** is the term used to describe solutions that involve extracting information from large volumes of often unstructured data to create a searchable knowledge store.

KNOWLEDGE MINING IN MICROSOFT AZURE



In this example, a travel web site uses Azure AI Search to power searching for information about destinations based on information extracted from images or text using AI services.

UNDERSTAND GENERATIVE AI

- Generative AI describes a category of capabilities within AI that create original content. People typically interact with generative AI that has been built into chat applications. Generative AI applications take in natural language input, and return appropriate responses in a variety of formats including natural language, image, code, and audio.

GENERATIVE AI IN MICROSOFT AZURE

In this example, an Azure OpenAI Service model is used to power a copilot application that can be used to generate original content in response to user *prompts*, such as a request to write a cover letter.

CHALLENGES AND RISKS WITH AI

Challenge or Risk	Example
Bias can affect results	A loan-approval model discriminates by gender due to bias in the data with which it was trained
Errors may cause harm	An autonomous vehicle experiences a system failure and causes a collision
Data could be exposed	A medical diagnostic bot is trained using sensitive patient data, which is stored insecurely
Solutions may not work for everyone	A home automation assistant provides no audio output for visually impaired users
Users must trust a complex system	An AI-based financial tool makes investment recommendations - what are they based on?
Who's liable for AI-driven decisions?	An innocent person is convicted of a crime based on evidence from facial recognition – who's responsible?

Responsible AI: Six Guiding Principles

Microsoft requires every Azure AI solution to be designed and governed against these principles:

Fairness

AI systems should treat all people fairly, without bias.

Reliability & Safety

Systems should perform consistently and safely as intended.

Privacy & Security

Protect data and respect user privacy at every stage.

Inclusiveness

Empower and engage people of all abilities and backgrounds.

Transparency

Systems should be understandable and their limitations clear.

Accountability

People remain responsible for how AI systems behave.

FAIRNESS

RELIABILITY AND SAFETY

PRIVACY AND SECURITY

INCLUSIVENESS

TRANSPARENCY

ACCOUNTABILITY

Microsoft Foundry: The Unified AI Platform

Unlike AI-900, which introduced individual Azure AI services one by one, AI-901 is built around Microsoft Foundry — the single hub for deploying models, building single or multi-agent solutions, and connecting AI capabilities into real applications.

Model Catalog

Discover and deploy foundation, open-source, and partner models.

Agent Building

Compose agents with memory, tools, and reasoning.

Governance

Apply Responsible AI controls and content safety centrally.

How AI Reaches an Application

Azure AI Service / Model

Microsoft Foundry (deploy, secure, govern)

REST API / SDK

Client Application or Agent

SUMMARY

- Summarize the main points of today's lecture